

Efficient Diffusion Language Models: A Comprehensive Survey

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GitHub Repository: <https://github.com/MessiX77/Awesome-Efficient-dLLMs>

Abstract

Diffusion language models (dLLMs) have recently emerged as a promising alternative to autoregressive (AR) language models, offering competitive performance with a fundamentally different generation paradigm. Instead of producing tokens strictly left-to-right, dLLMs iteratively refine partially masked sequences, enabling bidirectional context modeling and supporting parallel token updates. This design creates new opportunities for accelerating generation and improving controllability. However, practical deployment of dLLMs remains challenging due to the high cost of training at scale and the lack of mature inference optimizations such as cache-friendly decoding and robust parallel sampling. In this work, we provide a comprehensive overview of recent progress on *efficient* dLLMs. To reflect how efficiency bottlenecks arise across the full lifecycle of dLLMs, spanning training, decoding, and deployment, we organize existing approaches into five categories: *Training*, *Inference*, *Context*, *Framework*, and *Multimodality*. For each category, we summarize representative methods and highlight their motivations, key techniques, and empirical trade-offs. Finally, we discuss open challenges and outline promising directions for future research toward high-throughput, robust, and practically deployable diffusion language models.

1 Introduction

Autoregressive (AR) large language models (LLMs) have shown strong performance across tasks, but their left-to-right decoding generates tokens sequentially, limiting parallelism and making inference inefficient [Yang et al., 2025a]. Diffusion language models (dLLMs) offer an alternative paradigm by iteratively refining partially masked sequences [Gong et al., 2022; Song et al., 2025b]. This enables bidirectional context modeling and allows multiple tokens to be updated in parallel, making dLLMs a promising direction for efficient generation and improving controllability [Yang et al., 2025c; You et al., 2025; Zhu et al., 2025a; Liu et al., 2025a].

However, deploying dLLMs in practice still faces several efficiency bottlenecks. First, training dLLMs from scratch is highly data- and compute-intensive, and early open-source models (e.g., LLaDA [Nie et al., 2025]) remain less competitive than strong AR counterparts. Second, during inference, dLLMs have not yet shown consistent speed advantages over AR models with optimized serving engines like vLLM. This is largely due to the incompatibility of standard KV caching with bidirectional attention and the generation quality degradation of parallel decoding [Wu et al., 2025b]. Third, supporting long-context and variable-length generation is non-trivial, as many dLLMs rely on fixed denoising schedules and pre-specified output lengths.

In this survey, we provide the first comprehensive review of works that aim to improve the efficiency of dLLMs and enable their practical deployment. As illustrated in Figure 1, we organize the literature from multiple perspectives. For *training efficiency*, we summarize approaches that convert pre-trained AR models into dLLMs [Ye et al., 2025; Wu et al., 2025a] as well as architectural designs [Zhu et al., 2025b] that enable efficient dLLMs. For *inference efficiency*, we review parallel decoding strategies [Wu et al., 2025b; Chen et al., 2025b], dynamic KV-cache management [Ma et al., 2025a; Jiang et al., 2025], and compression techniques [Lin et al., 2025b; Agrawal et al., 2025] that improve the speed-quality trade-off. We further discuss *context scalability*, including methods for variable-length and long-context generation [Liu et al., 2025b; Arriola et al., 2025a]. In addition, we highlight emerging *frameworks* that support dLLM training, evaluation, and production serving [Zhou et al., 2025c; Ma et al., 2025b], and summarize recent progress on efficient *multimodal diffusion language models* [Yu et al., 2025b].

Although existing surveys [Li et al., 2025g; Yu et al., 2025a] have reviewed diffusion language models broadly, our survey focuses specifically on efficiency-related research. Our main contributions are threefold: (1) we propose a unified taxonomy that organizes efficient dLLM techniques into five categories, including *Training*, *Inference*, *Context*, *Framework*, and *Multimodality*; (2) we systematically review representative works in each category and analyze how they address key bottlenecks; and (3) we outline open challenges and promising future directions for efficient dLLMs.

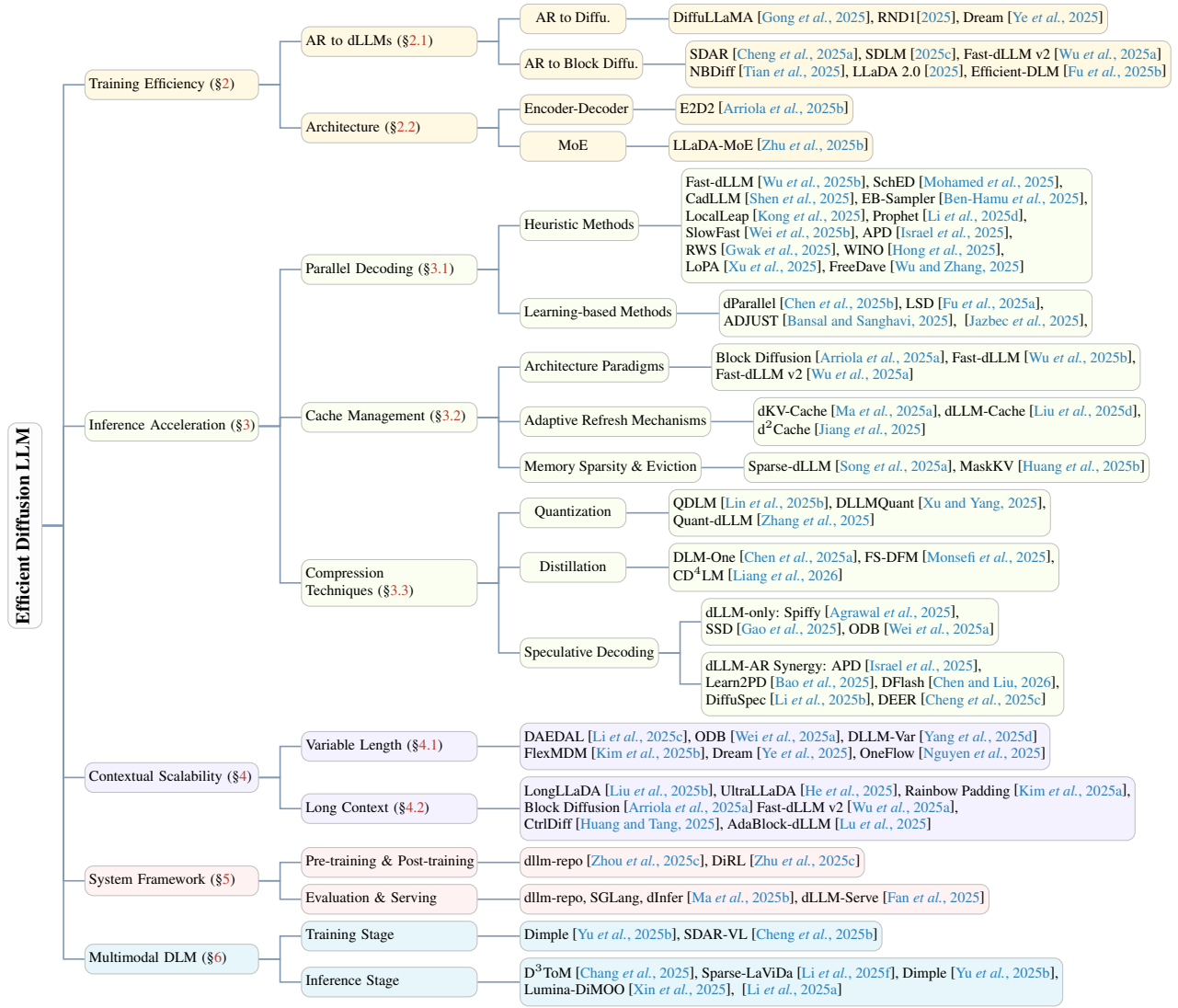


Figure 1: Our taxonomy tree of Efficient Diffusion Language Models (dLLMs).

2 Training Efficiency

2.1 AR to dLLMs

AR to Diffusion. Training diffusion language models from scratch at scale is computationally expensive. A practical alternative is to start from a pretrained AR model and focus on the core challenge of transition: how to convert an AR model into a diffusion-ready backbone without losing its linguistic competence. DiffuLLaMA [Gong et al., 2025] explicitly targets this AR-to-diffusion transfer, introducing dedicated transition techniques, such as attention-mask annealing, shift operations, and a time-embedding-free architecture. Dream-7B [Ye et al., 2025] pushes this direction further by turning the transition into a full training pipeline, notably incorporating context-adaptive token-level noise scheduling to improve downstream performance. In contrast, RND1 [Chandrasegaran et al., 2025] argues that the transition can be simplified: it directly switches from AR initialization to dLLM

training, avoiding explicit intermediate adaptation steps (e.g., attention annealing).

AR to Block Diffusion. Recent work on block diffusion language models [Arriola et al., 2025a], which generates text block-by-block while denoising the tokens within each block rather than denoising the entire sequence iteratively, shows that this approach preserves a structured dependency pattern more naturally aligned with AR pretraining. Subsequent studies indicate that adapting AR models to block-wise diffusion can be substantially more efficient than converting them to fully diffusion-based language models. SDAR [Cheng et al., 2025a] is the first to propose a highly efficient AR-to-block diffusion conversion pipeline. It leverages dense token-level supervision inherited from AR pretraining, enabling the diffusion objective to be aligned with only a short adaptation stage. Building on this efficiency-oriented view, SDLM [Liu et al., 2025c] further introduces a Next Sequence Prediction (NSP) objective that interleaves noise blocks with tar-

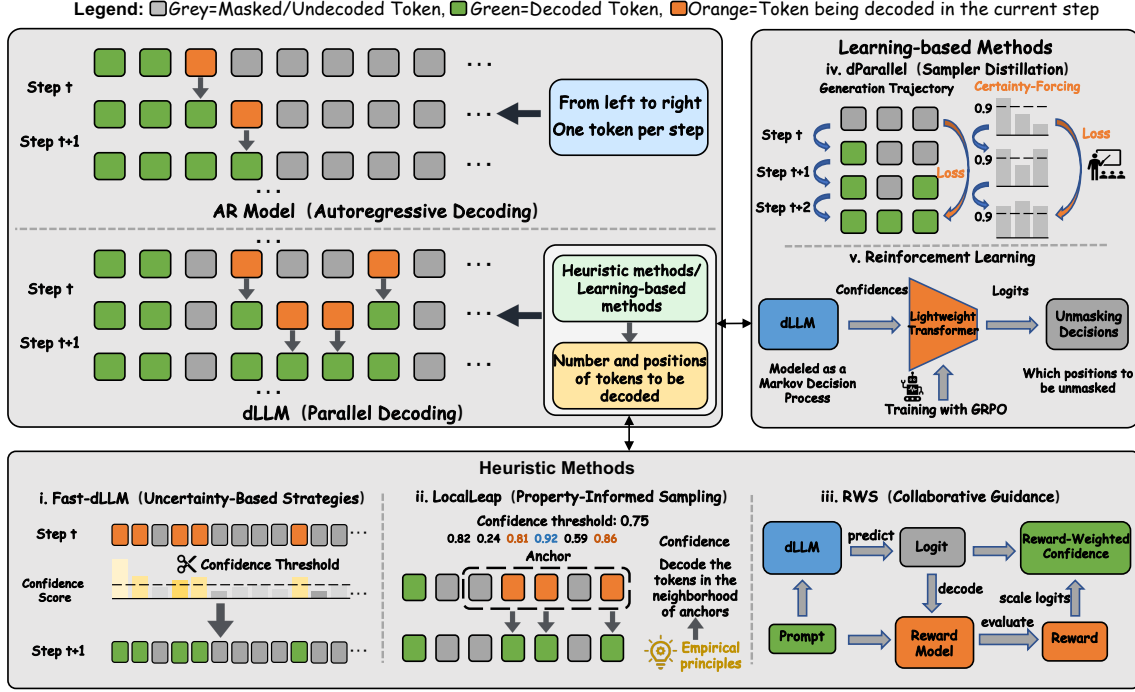


Figure 2: **Illustration of Parallel Decoding for dLLMs.** Unlike AR models that decode one token per step in a left-to-right manner, dLLMs can update multiple tokens simultaneously. In practice, heuristic or learning-based strategies are used to decide *which* token positions to decode and *how many* tokens to update at each step. Subfigures illustrate representative and state-of-the-art decoding methods.

get blocks, allowing parallel training while retaining block-level structure. Several subsequent works further refine the adaptation mechanism. NBDiff [Tian et al., 2025] designs a context-causal attention mask tailored for block diffusion adaptation and augments training with an auxiliary AR loss to improve convergence and knowledge retention. Fast-dLLM v2 [Wu et al., 2025a] identifies AR-friendly block-wise attention patterns and shows that effective block diffusion models can be obtained through lightweight post-training. Similarly, Efficient-DLM [Fu et al., 2025b] demonstrates that continuous pretraining with block-wise attention initialized from AR weights yields strong efficiency gains at low cost. More recently, LLaDA 2.0 [Bie et al., 2025] proposes a Warmup-Stable-Decay continual pretraining schedule to further stabilize the AR-to-block diffusion transition. Overall, these methods highlight that the core advantage of block diffusion lies in structurally aligning diffusion objectives with AR pre-training, making efficient adaptation possible without heavy architectural or training-stage interventions.

2.2 Architecture

Architecture choices in dLLMs are primarily motivated by efficiency considerations, as diffusion training and inference inherently involve repeated processing of partially noised sequences. Two design patterns are prevalent: decoupling clean and noisy representations to avoid redundant computation, and increasing model capacity through sparsity without proportional compute growth. E2D2 [Arriola et al., 2025b] adopts an **encoder-decoder** architecture for block diffusion, where the encoder processes clean inputs and the

decoder focuses on denoising, effectively reusing representations and reducing training cost by roughly half compared to decoder-only models of equal size. Orthogonally, **Mixture-of-Experts** (MoE) introduces sparse expert routing to scale model capacity efficiently. For instance, LLaDA-MoE [Zhu et al., 2025b] is the first to integrate MoE into diffusion language models, showing that sparse MoE-based dLLMs can outperform dense diffusion baselines and remain competitive with autoregressive models while activating substantially fewer parameters at inference time.

3 Inference Acceleration

3.1 Parallel Decoding

Currently, the open-source dLLMs have yet to fully achieve their high-throughput potential in real-world applications, primarily due to the performance degradation caused by decoding multiple tokens in parallel. To address this, several approaches have been proposed by improving the parallel decoding and sampling strategy, which can be categorized into two types: heuristic methods and learning-based methods.

Heuristic Methods leverage handcrafted metrics and empirical observations to optimize the decoding process, offering plug-and-play efficiency gains without extensive retraining. These approaches can be categorized into three dimensions:

- **Uncertainty-Based Strategies:** These methods leverage token-level uncertainty signals (e.g., confidence scores or entropy) to determine *which* tokens to decode and *how many* tokens to update at each step. Fast-dLLM [Wu et al., 2025b] selects tokens whose confidence exceeds a pre-

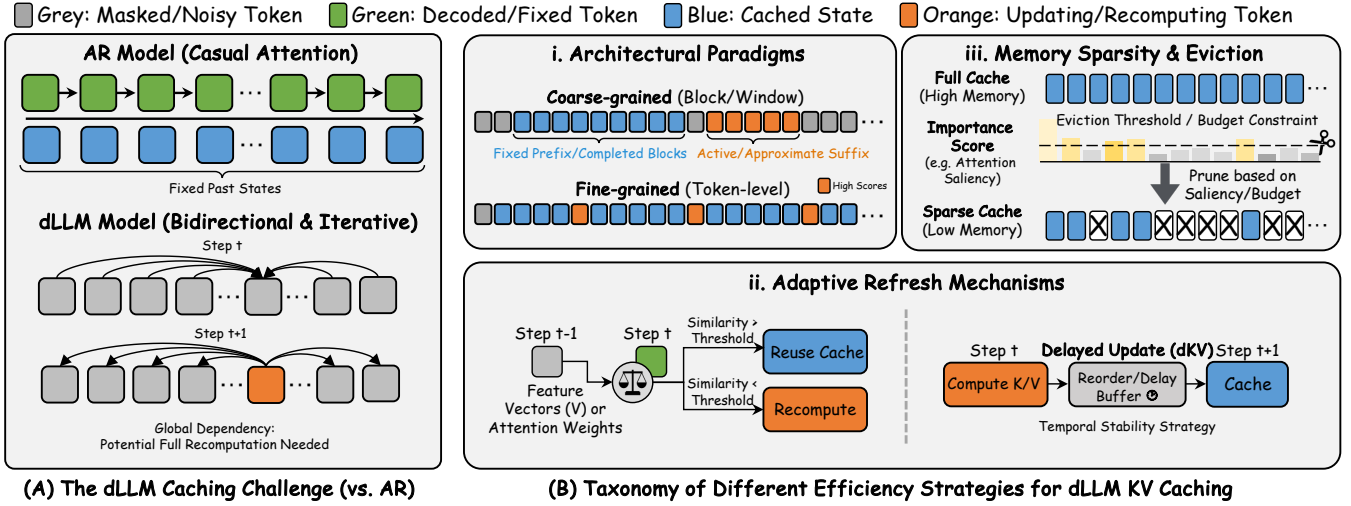


Figure 3: **Illustration of Cache Management for dLLMs.** (A) The key challenge in dLLM caching. Unlike AR models, dLLMs rely on bidirectional attention and iteratively refine the entire sequence, so KV states are not fixed and cannot be reused. (B) A taxonomy of efficiency strategies for dLLM KV caching: (i) *granularity control*, from coarse-grained blocks to fine-grained tokens; (ii) *adaptive refresh*, using similarity thresholds or delayed updates; and (iii) *memory sparsity and eviction*, guided by saliency signals and budget constraints.

set threshold, enabling safe and effective parallel decoding. SchED [Mohamed et al., 2025] proposes a training-free early-exit mechanism that halts decoding based on smooth, progress-dependent confidence thresholds, delivering substantial speedups. CadLLM [Shen et al., 2025] dynamically adjusts block size, step size, and unmasking thresholds based on confidence signals to improve throughput with competitive accuracy. EB-Sampler [Ben-Hamu et al., 2025] uses an entropy-bound scheduler to adaptively select the number and order of token updates per step, controlling both model and dependency errors.

- **Property-Informed Sampling:** Rather than relying on explicit uncertainty scores, these methods exploit empirical decoding patterns such as locality, progressive consistency, and early stabilization. LocalLeap [Kong et al., 2025] leverages local determinism and spatial consistency to perform parallel decoding within neighborhoods around high-confidence anchor tokens. Prophet [Li et al., 2025d] utilizes an early answer convergence phenomenon and terminates decoding once token transitions stabilize. Slow-Fast Sampling [Wei et al., 2025b] summarizes three principles for the decoding of dLLMs: certainty, convergence, and positional principle. Guided by these three principles, it designs a cyclical “exploration-acceleration” two-phase sampling strategy, significantly enhancing the inference efficiency of dLLMs.
- **Collaborative Guidance & Speculative Verification:** To further boost reliability, some methods introduce external guidance or internal verification loops. APD [Israel et al., 2025] and RWS [Gwak et al., 2025] utilize auxiliary autoregressive or reward models to refine the joint distribution and prioritize token decoding. On the architectural side, WINO [Hong et al., 2025], LoPA [Xu et al., 2025], and FreeDave [Wu and Zhang, 2025] adopt a speculative paradigm: they first draft multiple candidate tokens and

then use bidirectional context or internal self-verification to prune errors. This allows for revocable decoding, enabling the model to adaptively correct earlier stages as more contextual information becomes available.

Discussion. Heuristic methods offer strong practicality due to their simplicity and plug-and-play nature, while their effectiveness largely depends on the quality of uncertainty signals and the robustness of verification under aggressive settings.

Learning-based Methods generally encompass sampler distillation, trained auxiliary modules, and reinforcement learning, enabling significant inference acceleration in both semi-autoregressive and fully parallel scenarios:

- **Sampler Distillation:** dParallel [Chen et al., 2025b] designs a novel training strategy named certainty-forcing distillation, which distills the model to adhere to its original sampling trajectories and enables it to achieve high certainty over the masked tokens more rapidly and concurrently. This strategy significantly reduces the number of decoding steps while preserving the output quality of the model. LSD [Fu et al., 2025a] employs a distillation approach that enables a student sampler with fewer steps to learn and align with the intermediate score trajectories of a teacher sampler that utilizes more steps. Additionally, LSD+ further learns a non-uniform time schedule to adaptively allocate sampling steps.
- **Trained Auxiliary Modules:** ADJUST [Bansal and Sanghavi, 2025] trains a lightweight single-layer sampler based on an existing dLLM. Through several forward passes of this lightweight sampler after a single full-model forward pass, approximately sampling multiple tokens from the joint distribution can be achieved.
- **Reinforcement Learning:** [Jazbec et al., 2025] formulates dLLM sampling as a Markov Decision Process and use GRPO to train a lightweight policy network that adaptively selects token positions to unmask at each timestep.

Discussion. Overall, learning-based methods provide a systematic path to accelerate diffusion decoding by optimizing sampling policies directly, though they typically introduce extra training overhead and require careful design to maintain competitiveness across tasks and prompts.

3.2 Cache Management

The deployment of dLLMs is fundamentally hindered by the incompatibility of standard Key-Value (KV) caching with bidirectional attention mechanisms. Unlike AR LLMs, where past states remain fixed and cached KV pairs can be reused across decoding steps [Hao *et al.*, 2025b; Liu *et al.*, 2024], dLLMs iteratively refine the entire sequence and thus require repeated updates to token representations. As illustrated in Figure 3, current works have introduced caching mechanisms to dLLMs by architectural adaptations, algorithmic refinements, and memory optimizations.

Architecture Paradigms. Architecture approaches to caching in dLLMs mainly differ in the granularity at which generation dependencies are fixed and reused, as shown in Figure 3(i). Coarse-grained approaches, such as Block Diffusion [Arriola *et al.*, 2025a], adopt a window-based scheme that caches completed prefixes as fixed “Blue” blocks and restricts diffusion to an active “Orange” suffix. Fast-dLLM [Wu *et al.*, 2025b] and Fast-dLLM v2 [Wu *et al.*, 2025a] refine this idea with a DualCache design that separates exact prefix states from approximate masked suffix representations. In contrast, fine-grained strategies operate at the token level and selectively recompute only a small set of unstable tokens, which are identified using criteria such as confidence scores or attention weights, while directly reusing cached states for the remaining stable tokens.

Adaptive Refresh Mechanisms. For standard diffusion architectures, the main challenge in caching is KV drift, where token representations evolve across denoising steps, as shown in Figure 3(ii). Adaptive Refresh Mechanisms address this issue by dynamically deciding whether cached states should be reused or recomputed based on stability signals such as feature similarity or attention. dLLM-Cache [Liu *et al.*, 2025d] proposes a V-Verify strategy that compares Value vectors against a similarity threshold, reusing the cache when changes are small and recomputing otherwise. dKV-Cache [Ma *et al.*, 2025a] adopts a temporal stability strategy with delayed updates, postponing cache commits until token states stabilize. d^2 Cache [Jiang *et al.*, 2025] further refines this idea by prioritizing recomputation for high-attention groups while freezing low-attention states.

Memory Sparsity & Eviction. For long-context generation, cache management in dLLMs must operate under strict memory budgets, see Figure 3(iii). Memory sparsity and eviction methods [Nguyen-Tri *et al.*, 2025] address this by retaining only the most informative token states. Sparse-dLLM [Song *et al.*, 2025a] assigns attention-based importance scores to tokens and dynamically evicts low-importance entries, forming a sparse cache that preserves critical context. MaskKV [Huang *et al.*, 2025b] extends this idea with the *Mask as Prophet* principle, aggregating importance signals from mask tokens to guide more reliable eviction decisions.

By pruning based on global importance, these approaches cut memory footprint with minimal impact on denoising.

Discussion. These innovations mitigate the quadratic complexity of bidirectional attention, enabling dLLMs to approach the inference latency of AR models without sacrificing generation quality. Future research is expected to pivot from pure acceleration to broader applications, such as optimizing cache dynamics for multi-modal synthesis [Xin *et al.*, 2025] and utilizing the KV cache as a compressed protocol for multi-agent communication, thereby solidifying dLLMs as a scalable foundation for unified generative systems.

3.3 Compression Techniques

Quantization. To mitigate the substantial memory cost and computational overhead, Post-Training Quantization [Yang *et al.*, 2024; Yang *et al.*, 2025b; Lin *et al.*, 2024b] has been extended from AR LLMs to dLLMs. QDLM [Lin *et al.*, 2025b] establishes the first comprehensive benchmark, identifying distinct activation outliers in dLLMs and evaluating state-of-the-art PTQ baselines. Subsequent works address dLLM-specific features: DLLMQuant [Xu and Yang, 2025] introduces a fine-grained quantization scheme that accounts for temporal embedding, mask ratios, and attention distributions. Meanwhile, Quant-dLLM [Zhang *et al.*, 2025] identifies the critical role of timestep-dependent masking and proposes a data-aware, any-order quantizer, enabling extreme 2-bit weight-only quantization. Despite these algorithmic advances, two primary gaps remain: (1) a lack of specialized low-bit kernel support to translate theoretical compression into wall-clock speedup, (2) the unaddressed synergy between quantization and dLLM-specific caching mechanisms and (3) the need for Quantization-aware Training [Huang *et al.*, 2025a; Huang *et al.*, 2026] to better preserve quality.

Distillation. To bridge the sampling speed gap between dLLMs and AR LLMs, several works utilize distillation techniques to compress the iterative denoising process into minimal steps. DLM-One [Chen *et al.*, 2025a] focuses on optimizing continuous dLLMs and distilling a teacher’s score function into a one-step generator in continuous embedding space, achieving a $500\times$ speedup without iterative refinement. For discrete dLLM generation, FS-DFM [Monsefi *et al.*, 2025] introduces the few-step discrete flow-matching, achieving competitive long-text perplexity with only eight sampling steps, effectively balancing generation quality and throughput. Furthermore, CD⁴LM [Liang *et al.*, 2026] uses discrete-space consistency distillation to train the model to denoise correctly even when intermediate steps are skipped. Collectively, these approaches suggest that distillation-based methods provide a promising pathway for accelerating dLLM inference. However, preserving generation quality, especially long-form coherence, reasoning ability, and robustness, under aggressive step reduction remains a central challenge. Moreover, jointly exploring distillation with compression techniques such as pruning [Zhang *et al.*, 2024; Lin *et al.*, 2024a; Xing *et al.*, 2025; Hao *et al.*, 2025a] is still underexplored.

Speculative Decoding (SD). Speculative decoding is a family of inference acceleration techniques that improves

throughput by *decoupling proposal and validation*: the system first proposes multiple candidate token updates in parallel and then verifies them with a stronger decision rule to avoid quality degradation [Chen *et al.*, 2023; Li *et al.*, 2024]. While classical SD for autoregressive LLMs often relies on a separate lightweight draft model, recent dLLM work generalizes this paradigm into two directions: *dLLM-only speculation* and *dLLM-AR synergy*. Below, we organize existing methods under these two categories and summarize how they improve the throughput and quality.

dLLM-only Speculation optimizes the internal denoising process or uses dLLMs for both drafting and verification.

- **Self-Speculation:** This category focuses on unleashing the dLLM’s inherent potential to accelerate itself without external draft models. Spiffy [Agrawal *et al.*, 2025] introduces an “Auto-Speculation” mechanism, where the model predicts future intermediate states within its own denoising process. These predictions are structured into “Directed Draft Graphs” calibrated offline to maximize the acceptance rate during parallel verification. Similarly, SSD [Gao *et al.*, 2025] leverages the model to self-draft multiple tokens and verifies them using a “Hierarchical Verification Tree”, enabling the acceptance of valid sub-sequences in a single forward pass.
- **Iteration Optimization:** Addressing the memory-bound nature of decoding stage, ODB-dLLM [Wei *et al.*, 2025a] introduces a “Jump-Share” speculative decoding mechanism that selectively skips denoising steps (Jump) and shares KV caches across layers (Share). This strategy optimizes arithmetic intensity, significantly reducing memory bandwidth pressure during iterative denoising phase.

dLLM-AR Synergy Speculation bridges dLLM and AR models to leverage their complementary strengths.

- **Adaptive Speculation:** Motivated by the fact that token difficulty varies across contexts, these methods adapt the speculation budget by dynamically deciding how many tokens to draft in parallel and which drafts require further refinement. APD [Israel *et al.*, 2025] leverages a small auxiliary AR model to estimate the joint plausibility of dLLM drafts and adjusts the parallel lookahead accordingly, improving the throughput–quality trade-off. Learn2PD [Bao *et al.*, 2025] takes a learning-based route by training a lightweight filter model to predict draft correctness; low-confidence tokens are selectively re-masked and re-denoised, enabling selective verification and early acceptance for “easy” tokens.
- **Hybrid Verifier Architectures:** These approaches marry the parallel generation capabilities of Diffusion (as drafters) with the precise reasoning of Autoregression (as verifiers). DEER [Cheng *et al.*, 2025c] addresses the “uncertainty accumulation” typical of AR drafters by employing a discrete dLLM to generate high-quality block drafts, which are subsequently verified by the target AR model. DiffuSpec [Li *et al.*, 2025b] proposes a training-free solution using “Causal-Consistency Path Search” (CPS) to extract AR-aligned sequences directly from the dLLM’s internal token lattice. Furthermore, DFlash [Chen and Liu, 2026] enhances drafting by conditioning a lightweight diffusion

model on the “hidden context features” of the large target AR model, bridging the representation gap between the compact drafter and the large verifier.

Discussion. Looking forward, effective SD algorithms for dLLMs will likely require adaptive speculation strategies that are tightly coupled with diffusion-specific decoding dynamics, as well as synergistic integration with model compression techniques. Such co-design is essential to fully exploit hardware parallelism while preserving generation quality, particularly under increasingly aggressive acceleration regimes.

4 Contextual Scalability

4.1 Variable Length

Early diffusion language models rely on a fixed, pre-specified output length, which limits practical usability and wastes computation by denoising padding tokens or over-allocated sequence slots [Han *et al.*, 2023]. Enabling native variable-length generation is therefore essential for efficient dLLMs, as it allows computation to adapt to the actual response length while preserving diffusion’s parallel decoding advantages.

Inference-Time Adaptive Length Control. This line of work determines output length on the fly during inference using confidence-based stopping signals, rather than fixing it in advance. DAEDAL [Li *et al.*, 2025c] starts from a short sequence, adjusts length based on End-Of-Sequence (EOS) token confidence, and then inserts masks at low-confidence positions for iterative refinement. dLLM-Var [Yang *et al.*, 2025d] enforces explicit EOS modeling by always masking the EOS token during training and conditionally appending new blocks at inference when no EOS is produced, enabling open-ended generation. Complementarily, ODB-dLLM [Wei *et al.*, 2025a] introduces adaptive length prediction to terminate decoding at the first high-confidence EOS.

Dynamic-Canvas and Any-Order Generation. This paradigm frames generation as iterative refinement on a dynamic canvas, where length emerges implicitly from insertion and editing rather than explicit length control. FlexMDM [Kim *et al.*, 2025b] decouples length from token prediction by first estimating insertion counts between existing positions and then filling the inserted masks via posterior denoising, enabling flexible-length generation under non-causal diffusion. Dream [Ye *et al.*, 2025] combines autoregressive initialization with diffusion-based infilling to support the generation and completion of text segments with arbitrary lengths. Extending this idea further, OneFlow [Nguyen *et al.*, 2025] generalizes dynamic-canvas refinement to multimodal through an edit-flow mechanism, enabling parallel, interleaved synthesis of variable-length text and image content without fixed-length allocation.

Discussion. These approaches make variable-length generation practical for efficient dLLMs in real-world settings. Future work should develop more robust length-adaptive decoding to maintain stable termination and coherence.

4.2 Long Context

Despite significant progress in AR LLM for long-context processing, dLLMs have historically struggled with context scal-

ability due to fundamental architectural mismatches [Dat et al., 2024]. Recent advances have enabled context windows to expand from thousands to over 128K tokens while preserving generation quality, primarily through following techniques:

- **Position Encoding & Extrapolation:** LongLLaDA [Liu et al., 2025b] presents the first systematic analysis of long-context extrapolation in dLLMs, showing that dLLMs degrade more gracefully beyond their training context and retain strong sensitivity to recent context segments. Based on this, it derives a training-free RoPE scaling rule by computing the critical dimension and scaling factor, enabling effective context extension without parameter modification. UltraLLaDA [He et al., 2025] further extends the context window to 128K tokens through diffusion-aware NTK position encoding and a multi-document concatenation masking strategy during post-training.
- **Blockwise Diffusion:** To enable long-context generation without prohibitive computation, blockwise diffusion decomposes extended sequences into segments and performs parallel denoising within blocks while coordinating dependencies across blocks. Block Diffusion [Arriola et al., 2025a] combines sequential block generation with intra-block parallel refinement to maintain coherence at scale, while Fast-dLLM v2 [Wu et al., 2025a] further improves long-context efficiency through bidirectional block modeling and hierarchical caching. CtrlDiff [Huang and Tang, 2025] and AdaBlock-dLLM [Lu et al., 2025] adapt block granularity during generation based on semantic or attention cues, allocating computation where long-range dependencies are most critical.
- **Memory optimization:** These techniques mitigate the quadratic attention bottleneck in long-sequence generation by reducing memory overhead. Rainbow Padding [Kim et al., 2025a] tackles early termination in instruction-tuned dLLMs through cyclic multi-token padding patterns that prevent premature EOS prediction.

Discussion. Together, these advances establish a promising foundation for long-context dLLMs. However, fully realizing their practical benefits will require deeper co-design across acceleration techniques to maintain global coherence and superior performance under extreme context lengths.

5 System Framework

As diffusion language models move toward large-scale deployment, the focus has shifted from algorithmic feasibility to system-level efficiency and scalability. While vLLM [Kwon et al., 2023] and SGLang [Zheng et al., 2024] are widely utilized as industry standards for autoregressive LLM serving, the specialized ecosystem for dLLMs is still in its nascent stages and requires broader community support to match this maturity. Current progress indicates a trend toward integrating dLLM support into existing high-performance engines and developing dedicated frameworks:

- **Unified Training and Evaluation:** dllm-repo [Zhou et al., 2025c] provides a comprehensive, research-oriented library that standardizes the full diffusion LLM lifecycle, including training, fine-tuning, and evaluation. It supports LoRA,

DeepSpeed, and FSDP, while providing reference implementations for training algorithms such as masked diffusion, block diffusion, and edit flows. [Peng et al., 2025] benchmark the throughput of dLLMs against AR models and provide insights into practical acceleration strategies for dLLMs.

- **Post-training Optimization:** DiRL [Zhu et al., 2025c] addresses the critical training-inference mismatch in dLLMs by establishing an efficient post-training framework that supports scalable reinforcement learning via online policy updates and training-inference co-design.
- **Mainstream Serving Integration:** SGLang has recently introduced initial support for the LLaDA 2.0 series models. SGLang leverages its existing Chunked-Prefill pipeline to implement computational support for Block Diffusion LLM, enabling high-performance deployment of 16B and 100B parameter diffusion models.
- **Specialized Inference Engines:** dInfer [Ma et al., 2025b] develops a modular inference pipeline that decouples model logic from diffusion iteration management, achieving a 10 $\times$ speedup over baseline frameworks through the integration of CUDA Graphs and loop unrolling.
- **Production Serving:** To tackle the highly variable memory footprint of dLLM inference in production GPUs, dLLM-Serve [Fan et al., 2025] employs Logit-Aware Activation Budgeting and a Phase-Multiplexed Scheduler to manage the massive memory overhead of logit tensors, effectively enhancing throughput in resource-constrained scenarios.

Discussion. Collectively, these frameworks form a growing ecosystem that addresses the full lifecycle of dLLM deployment, from optimized single-request latency to high-concurrency production serving and efficient alignment. However, as dLLMs move toward more complex non-autoregressive architectures and multimodal settings, there is a pressing need for specialized open-source frameworks that provide production-level robustness, portable performance optimizations across different hardware backends, and seamless integration with existing serving infrastructures.

6 Multimodal DLM Efficiency

Training Stage. To alleviate the inherent token-by-token decoding latency of autoregressive multimodal models [Zhou et al., 2025a; Zhou et al., 2025b; Lin et al., 2025a], recent work has explored Multimodal Diffusion Language Model (MDLM) as a parallel and more controllable generation paradigm. Dimple [Yu et al., 2025b] develops a MDLM by employing a hybrid “autoregressive-then-diffusion” training paradigm to stabilize convergence. Pushing block-wise efficiency, SDAR-VL [Cheng et al., 2025b] introduces an asynchronous noise scheduling strategy, demonstrating that block-wise discrete diffusion can outperform strong autoregressive baselines in vision-language understanding tasks.

Inference Stage. To address the computational intensity of multimodal denoising, recent works propose acceleration from algorithmic perspectives. D³ToM [Chang et al., 2025] dynamically merges redundant visual tokens via “decider-guided” importance maps, while Sparse-LaViDa [Li et al.,

2025f] truncates unnecessary masked tokens by utilizing specialized register tokens for compact representation. Complementing these, [Li et al., 2025a] identifies a progressive information recovery mechanism in discrete diffusion, providing a theoretical basis for aggressive token pruning without sacrificing fidelity. Furthermore, Dimple optimizes the inference pipeline through “confident decoding,” which dynamically adjusts the number of tokens generated per step to reduce iterations by two-thirds, and re-incorporates the prefilling mechanism from AR models to jointly minimize generation complexity. Lumina-DiMOO [Xin et al., 2025] applies dLLM efficiency to multi-modal generation using a “Max-Logit” cache, freezing representations of tokens with high prediction confidence to accelerate image-text synthesis.

Discussion. Together, these advancements transition multimodal dLLMs from theoretical exploration to high-performance, sparse-inference systems. However, the rapid development of MDLM still necessitates multi-level optimizations, spanning from data-centric noise scheduling to hardware-aware kernel designs, to fully unlock their potential in real-time, large-scale multimodal applications. This remains a fertile ground for future research into unified, cross-modal diffusion architectures.

7 Future Research Directions

Training Efficiency. Future work should improve the data efficiency of AR-to-diffusion transfer and move beyond naive initialization. Designing curricula and data structures that fully exploit bidirectional conditioning, especially data with strong long-range dependencies, may help dLLMs surpass AR models in global reasoning. Moreover, training-inference co-design remains crucial: incorporating efficient decoding structures (e.g., learned schedules, block-wise refinement, cache-friendly attention) directly into training could significantly reduce the need for serving stage optimization.

Inference Optimization. A key challenge still lies in developing a unified paradigm for the throughput-quality trade-off in parallel decoding, together with diffusion-aware cache management. Unlike AR serving, which benefits from mature primitives such as PageAttention and FlashAttention, dLLMs still lack standardized abstractions for parallel sampling and KV cache. Another important direction is hardware-aligned compression and kernel optimization. Attaining wall-clock speedups comparable to AR engines requires a full-stack orchestration of algorithmic parallelization, system-level memory management, and customized Triton/CUDA kernels.

Context and Framework. Bridging the context scalability gap between dLLMs and modern AR models (which can reach 1M-token windows) requires mitigating the quadratic cost of bidirectional attention while preserving global coherence under parallel denoising. Blockwise diffusion is a promising direction, but it still requires more effective memory management and stable decoding strategies to remain efficient as sequence lengths continue to grow. In addition, the dLLM ecosystem still lacks mature open-source frameworks covering pretraining, post-training, and production serving. Building robust, reusable tooling will be essential for making dLLMs practically deployable in real-world applications.

Multimodal DLM. Discrete diffusion-based models provide a unified generation paradigm that naturally extends from text to multimodal understanding and generation [Luo et al., 2025; Li et al., 2025e; Shi et al., 2025]. A critical frontier involves enhancing spatial-semantic alignment in unified MDLMs to support high-fidelity perception tasks, such as document parsing and precise visual grounding. Reinforcement learning is also promising, as discrete tokens enable token-level rewards to better optimize quality and controllability [Yang et al., 2025c; Xin et al., 2025].

8 Conclusion

This survey provides a comprehensive overview of existing research efforts toward improving the efficiency of dLLMs. To capture efficiency bottlenecks across the dLLM pipeline, from pretraining and decoding to long-context scalability and multimodal extension, we organize the literature into five categories, summarize representative methods, and discuss their practical implications. Finally, we highlight open challenges and future directions, with the hope of inspiring further progress toward efficient and widely deployable dLLMs.

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